

```

SELECT Customers.CustomerID, Customers.CustFirstName || ' ' || Customers.CustLastName,
       StyleName, (SELECT COUNT(*)
                   FROM Musical_Preferences AS MP
                   WHERE MP.CustomerID = Customers.CustomerID)
       AS Preferences
FROM Customers INNER JOIN Musical_Preferences
ON Customers.CustomerID = Musical_Preferences.CustomerID
INNER JOIN Musical_Styles
ON Musical_Styles.StyleID = Musical_Preferences.StyleID
;

```

--The above query can be also be performed in this manner:

```

SELECT C.CustomerID, C.CustFirstName || ' ' || C.CustLastName
       AS Customer,
       COUNT(*) AS Preferences
FROM Customers AS C
       INNER JOIN Musical_Preferences AS MP
       ON MP.CustomerID = C.CustomerID
GROUP BY C.CustFirstName, C.CustLastName, C.CustomerID
;

```

-- The key difference between GROUP BY and OVER() is that GROUP BY applies aggregations across the
-- entire query, rolling-up the specified non-aggregate fields, thus usually reducing the number of
-- returned fields.

-- With OVER(), the same number for rows will be returned as the base query. Any aggregations will be
-- returned for each row in the range identified in the OVER() clause.

-- Predicates used: PARTITION BY, ORDER BY, ROWS (or RANGE)

```
SELECT C.CustomerID,
       C.CustFirstName || ' ' || C.CustLastName AS Customer,
       MS.StyleName,
       COUNT(*) OVER(
           PARTITION BY C.CustomerID)
       AS Preferences
FROM Customers AS C
      INNER JOIN Musical_Preferences AS MP
            ON MP.CustomerID = C.CustomerID
      INNER JOIN Musical_Styles AS MS
            ON MS.StyleID = MP.StyleID
;
```

Partial Output:

```

SELECT C.CustomerID,
       C.CustFirstName || ' ' || C.CustLastName AS Customer,
       MS.StyleName,
       COUNT(*) OVER(
           PARTITION BY C.CustomerID)
       AS Preferences
FROM Customers AS C
INNER JOIN Musical_Preferences AS MP
    ON MP.CustomerID = C.CustomerID
INNER JOIN Musical_Styles AS MS
    ON MS.StyleID = MP.StyleID
;

```

Output Messages Notifications

customerid integer	customer text	stylename character varying (75)	preferences bigint
10001	Doris Hartwig	Contemporary	2
10001	Doris Hartwig	Top 40 Hits	2
10002	Deb Waldal	60's Music	2
10002	Deb Waldal	Classic Rock & Roll	2
10003	Peter Brehm	Motown	2
10003	Peter Brehm	Rhythm and Blues	2
10004	Dean McCrae	Jazz	2
10004	Dean McCrae	Standards	2
10005	Elizabeth Hallmark	Classical	2

Another Example:

Query Query History

```
1 -- "For each customer, show the musical preference styles
2 -- they've selected. Show me a running total of the number of styles selected for
3 -- all customers."
4
5 SELECT C.CustomerID,
6        C.CustFirstName || ' ' || C.CustLastName AS Customer,
7        MS.StyleName,
8        COUNT(*) OVER (
9            ORDER BY C.CustomerID)
10       AS Preferences
11 FROM Customers AS C
12      INNER JOIN Musical_Preferences AS MP
13      ON MP.CustomerID = C.CustomerID
14      INNER JOIN Musical_Styles AS MS
15      ON MS.StyleID = MP.StyleID
16 ;
```

Data Output Messages Notifications



Showing rows: 1 to 36 Page N

	customerid integer	customer text	stylename character varying (75)	preferences bigint
1	10001	Doris Hartwig	Contemporary	2
2	10001	Doris Hartwig	Top 40 Hits	2
3	10002	Deb Waldal	60's Music	4
4	10002	Deb Waldal	Classic Rock & Roll	4
5	10003	Peter Brehm	Motown	6
6	10003	Peter Brehm	Rhythm and Blues	6
7	10004	Dean McCrae	Jazz	8
8	10004	Dean McCrae	Standards	8